Low cost dipstick immunoblot assay development for bovine pregnancy diagnosis

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Abstract

Low cost, affordable early pregnancy diagnosis kit is very deserving for livestock management and dairy business. This study proposes to develop a kit based on enzyme immunoassay (EIA). New Zealand white rabbits primed with 11-α-OH progesterone hemisuccinate-BSA conjugate were boosted with progesterone antigen for crude antisera production. Progesterone antibody was purified using an immunoaffinity chromatography column packed with 6- amino hexanoic acid sepharose 4B beads conjugated with progesterone carboxymethyl oxime and coated on nitrocellulose paper dipsticks. The 11-α-OH progesterone hemisuccinate-penicillinase conjugate solution was purified using Sephadex G-50 size exclusion column. All conjugates were prepared using carbodiimide reaction. Optimum pH and temperature for penicillinase enzyme highest activity and iodine concentration for starch iodine substrate solution were determined. Checker board analysis and titration were performed to optimize antibody, conjugate and milk dilutions for kit. Further kit was validated with randomly selected 20 cow milk samples for pregnancy diagnosis. Chromatography data showed peaks for fractions containing highly purified progesterone antibody and progesterone penicillinase conjugate respectively. Optimum dilutions of antibody, conjugate and milk samples were 1:20, 1:40, 1:3 (or 1:4) respectively to differentiate between 1 (non pregnant) and 5 (pregnant) ng/ml of progesterone concentration. Standard conditions of kit optimum function are pH 7.0 and 35 ºC. Penicillin solution was stable at 4 ºC for 3 weeks. Excellent correlation was observed for pregnancy diagnosed by developed kit to detect progesterone concentration in 20 milk samples and conventional rectal palpation method. Developed low cost dipstick EIA kit can be used for early pregnancy diagnosis successfully.

Keywords

Low cost, dipstick immunoblot assay, EIA kit bovine pregnancy diagnosis, diagonostic kit

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Introduction

Pregnancy diagnosis in bovine at early stage after insemination is very deserving for livestock management in dairy business (Balhara et al., 2013). Conventional rectal palpation method to detect pregnancy is cumbersome. It can deliver results if it is carried out at least 45 days after insemination. It also requires services of a experienced and trained veterinarian (Romano et al., 2007). Hence, It has been tried to identify various chemical indicators of pregnancy in body fluids, for example, tissue extracts, sweat, saliva, blood and milk etc (Volkery et al., 2012a). Among these progestrone had proved most reliable for early detection of pregnancy (Martinez et al., 1995; Sasser and Ruder, 1987). Progesterone is a steroid hormone and is produced by corpus luteum of ovary and adrenal cortex. Bovines follow reproductive cycle or estrous cycle which varies between 18- 25 days depending on bovine's species. These animals exhibit typical estrous symptoms at day zero when circulatory progesterone level is around 1 ng/ml. But with development of corpus luteum which secretes progesterone, there is a constant increase in circulatory level of this hormone. And this is maintained in pregnant animals. However, in non pregnant animals, corpus luteum regresses, consequently circulatory progesterone level also decreases. Therefore, progesterone level estimation become a prominent indicator to know bovine pregnancy status (Forde et al., 2011; Sartori and Barros, 2011).

Diagnostic kit development to detect efficiently quantitative and qualitative changes at very tiny scale is very worth for perspective of biotechnological endeavors. There have been reported many techniques to detect progesterone level; bioassays, fluorescence assay, double isotope procedure, gas- liquid chromatography, enzyme immunoassay (EIA), dipstick EIA, radioimmunoassay (RIA) are few of them (Bayemi et al., 2007; Rasbech et al., 1986; Volkery et al., 2012b) . Among all these techniques, dipstick EIA seems worthful because it does not need trained personal, creates health hazards (radioactive isotope exposure) and can detect easily progesterone concentration in milk samples. Milk samples are easily available and previous results showed that milk, in comparison to blood, contains much higher concentration of progesterone.

EIA kits of pregnancy diagnosis are available in Europe, USA, Canada and Australia. Most of them are very costly and unaffordable for farmers of many countries such as Indian farmers. For Indian perspective, a kit should be cheap, insensitive to temperature and storage conditions. In this regard, a dipstick EIA kit was developed using progesterone- penicillinase conjugate as an enzyme marker and starch- iodine-penicillin V as a substrate. This kit was further refined and improved for good accuracy. Standardization of various reagents is essential for refinement of a test. Hence it was performed; 1. Standardization various reagents is essential for refinement of a various reagents is essential for refinement of a test. Hence it was performed; 1. Standardization of functional ratio of concentrations of penicillin, starch and iodine. 2. Effect of temperature, pH and storage time on the conjugate activity. 3. Optimization of conjugate and progesterone antibody dilutions for dip-strip assay.

Materials and methods

Experimental animals

Post-natal castrated New Zealand white rabbits were maintained at the Animal House of College of Veterinary Sciences of the G.B. Pant University and used for the production of progesterone antisera. Milk samples were collected from the University Livestock Research Center and Veterinary Clinic of the College.

Progesterone antisera production

New Zealand white rabbits were primed with 11-α-OH progesterone hemisuccinate-bovine serum albumin (BSA) conjugates (Steraloids Inc., Witton) and then boosted with antigen for production of progesterone antisera. For this purpose, 500 microgram (µg) of 11-α-OH progesterone hemisuccinate-BSA conjugate was dissolved in 3 milliliter (ml) of 0.85%

Table 1 Immunization and bleeding schedule.

Day	Immunogen dose total	Adjuvant used	Remarks
Primary immunization Day 0	500 microgram	FCA; immunogen in 0.85% Normal saline solution	Multiple intradermal injection at 40-42 sites
I. Booster dose Day 45	100 microgram	FIA	Intradermal injection at 2-4 sites
II. Booster dose Day 55	100 microgram	FIA	Intradermal injections at 2-4 sites
III. Booster dose Day 70	100 microgram	FIA	Intradermal injections at 2-4 sites
Final bleeding Day 80			

sodium chloride solution and was drawn in 5 ml glass syringe. Three ml of Freund´s complete adjuvant (FCA; DIFCO, Detroit, MI) was drawn into another 5 ml glass syringe. These two syringes were interconnected by an adapter tube. Contents of both syringes were mixed thoroughly. Tube was kept in a beaker of ice considering as high temperature could lead to denaturation of antigen. This emulsion was for primary immunization. For booster immunization a similar emulsion was prepared using 100 µg of antigen and 3 ml of Freund´s incomplete adjuvant (FIA).

Rabbits were immunized on dorsal side on both sides of vertebral column, intradermally, at about 40-42 sites. Booster doses were injected in similar manner at about 2-4 sites. Immunization and bleeding schedule is presented in table 1.

Hundred micro liters (µl) of 15% levamisole (BAIF, India) was injected subcutaneously in each rabbit one day before primary immunization.

Two ml of blood samples about 12 days after last booster injection was collected from marginal ear vein of rabbit and crude serum was separated. Protein detection in this anti-progesterone serum was determined using Lowry´s method.

Affinity chromatography for progesterone antibody purification

Immunoaffinity chromatography was used to purify progesterone antibodies from crude serum. A chromatography column was packed with a affinity matrix of 6-amino hexanoic acid-sepharose 4B beads conjugated with progesterone carboxymethyl oxime (Sigma-Aldrich). This conjugation of beads with progesterone was performed using carbodiimide reaction method. In brief, 5 gram of 6-amino hexanoic acid sepharose 4B was soaked in excess of 0.1 M sodium chloride. Then it was washed with 0.5 M sodium chloride solution and subsequently with distilled water. Ten milligram (mg) of progesterone carboxymethyl oxime dissolved in 10 ml of dioxane was added to above slurry and pH was adjusted to 4.5-6.0. This mixture was stirred gently at room temperature and 1-ethyl 3-(3-dimethy amino propyl) carbodiimide (Sigma-Aldrich) was slowly added to a final concentration of 0.1 M. The pH was maintained at 4.5-6.0. Then mixture was kept on shaker and gently rotated and incubated for 24 hours at room temperature. Prepared slurry was now packed in a glass column and washed with 50% (v/v) aqueous 1, 4 dioxane till no progesterone was detected in wash. Column was stored in 0.15 M PBS, pH 7.2 containing azide until use.

Table 2 Protein loaded and recovered from affinity column chromatography.

Rabbit No.	Protein			
	Loaded	Recovered (mg)		Non specific
	(mg/ml)	pH 4.0	pH 2.7	(mg)
P7	280	2.130	2.170	276.70
P5	192.5	0.954	1.130	190.414

One ml of crude antiserum was diluted to 10 ml with 0.15 M PBS pH 7.2 and then loaded in affinity column. Undesired proteins were washed out till <0.1 absorbance was observed at 280 nm. For specific elution of progesterone antibodies, fractions of 1 ml each were eluted with 0.1 M glycine HCL gradient buffer pH 4.0 and pH 2.7 respectively. Following protein detection in elute by Lowry´s method, fractions of elute were collected in 0.5 M Tris buffer pH 8.5. Fractions with highest elution peak were collected and protein concentration was determined.

Antibody coated dipstick preparation

Dipsticks with nitrocellulose paper pad (NCP dipstick) were obtained from Advance Micro devices, Ambala, India and coated with affinity column purified antibody. Serial dilution of antibody was done in a coating buffer (Carbonate buffer, 0.05 M, pH 9.6) and 2 µl of diluted antibody was dotted on NCP dipsticks. Dipsticks were dried and washed with washing buffer (PBS-T; 0.1 M PBS, pH 7.4 containing 0.05% Tween-20). Sticks were then kept in blocking solution (2% Gelatin in PBS) for 2-4 hours at 37 ºC and washed again with washing buffer. Following, dipsticks were dried in dust free environment and stored at 4 ºC inside moisture proof plastic bags for further use.

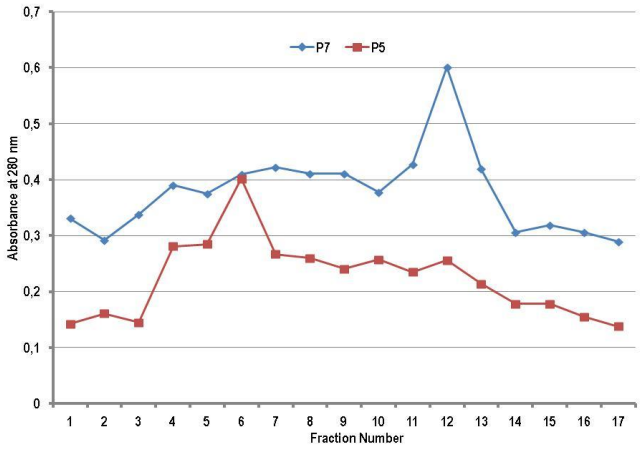


Figure 1 Elution profile of progesterone antisera (Rabbit Number P7, P5) from sepharose 4 B amino hexanoic acid: progesterone affinity column with 0.1 M Tris HCL buffer pH 4.0.

Progesterone penicillinase conjugate preparation

11-α-OH progesterone hemisuccinate was coupled covalently to penicillinase (Hindustan Antibiotics Ltd, Pune, India) enzyme by carbodiimide reaction. In brief, five mg of 11-α- OH progesterone hemisuccinate was dissolved in 100 µl of dioxane. Two hundred µl of 0.15 M PBS pH 7.2 was added to solution and pH was maintained between 4.5-6.0. Fifteen mg of 1-ethyl 3-(3-dimethy amino propyl) carbodiimide was added directly to solution maintaining pH between 4.5-6.0. Resultant mixture was incubated at room temperature (26-28 ºC) for 1 hour with intermittent shaking. Ten mg of penicillinase enzyme was dissolved in 300 µl of 0.15 M PBS pH 7.2 and was added drop-wise to resultant mixture. Mixture was then kept at 4 ºC for 16-18 hours for incubation. Final pH was maintained at 4.5. After, incubation mixture was loaded on G-50 column (1.5 cm x 30 cm), fractions with highest enzyme activity were collected.

Starch iodine substrate solution preparation

Two hundred mg of soluble starch in 20 ml distilled water was heated to a homogeneous mixture. 1.1 ml of starch solution in 10 ml of 0.1 M PBS pH 7.0 and 5.5 µl of iodine solution (0.08 M iodine in 3.2 M KI solution) was mixed with gentle shaking. Finally 10 µl of 1 mM Benzylpenicillin solution prepared in distilled water was added just prior to use.

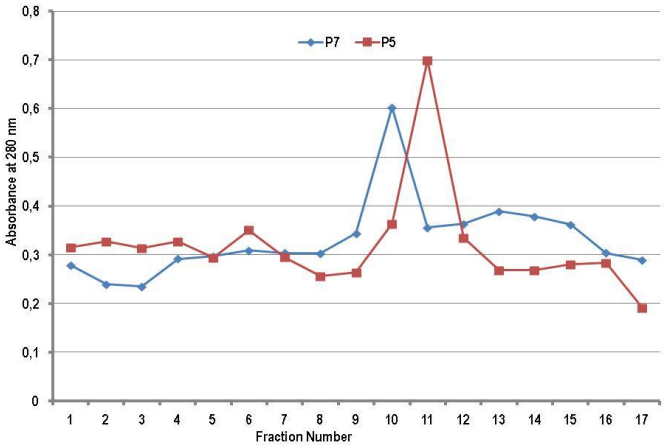


Figure 2 Elution profile of progesterone antisera (Rabbit Number P7, P5) from sepharose 4 B amino hexanoic acid: progesterone affinity column with 0.1 M Tris HCL buffer pH 2.7.

Experimental results

Progesterone antiserum purification

Affinity chromatography technique was used to purify antiserum collected from each animal as described in methods and materials. Protein concentration of antiserum was found 192.5 mg/ml (Rabbit no. P5) and 280.0 mg/ml (Rabbit no. P7). Diluted protein samples were loaded onto affinity columns. After removing nonspecific proteins with 0.15 M PBS buffer pH 7.2, thirty 1 ml fractions were collected in 0.5 M Tris- HCL buffer, pH 8.6, following elution with pH 4.0 and pH 2.7, 0.1 M Glycine-HCL buffer. Tris buffer was used as collection buffer to neutralize acid present in eluent. Elution profile of progesterone antibody at pH 4.0 and pH 2.7 are shown in figures 1 and 2. Total amount of protein recovered is shown in table 2.

Dipstick enzyme immunoassay

Affinity purified progesterone antibody and progesterone-penicillinase conjugate are used for this purpose. Standardization of various reagents was essential for refinement of test.

Starch iodide complex standardization

In iodometric analysis of penicillinase enzyme, starch-iodide and benzyl penicillin were used. Penicillinase catalyses the conversion of benzyl penicillin to penicilloic acid which in turn reduces iodide of starch-iodide resulting into decolourization of solution. Almost 4 micromole (µM) of iodide was reduced by 1 µM of penicilloic acid, a product of benzyl penicillin after enzyme catalysis. Keeping this in mind, different concentrations of iodine were analyzed with 10 µM of benzyl penicillin and 2% starch. Results in table 3 indicate that as iodine concentration increased, there was a marked increase in time of decolourization. It was observed that concentration in range of 40-50 µM of iodine was decolorized in a reasonable time of 7-15 minutes with 0.04 units of enzyme.

When starch iodide complex was scanned in wavelength range of 400-700 nm for maximum absorption, results indicated a peak in range of 562-565 for spectrophotometric analysis of complex as shown in figure 3. Absorption between 560-570 nm was selected.

Table 3 Effect of iodine concentration on substrate decolourization. Experiment designed in triplicate.

Serial no.	Iodine concentration (micromole)	Decolourization time (minute)
1	40	7
2	50	15
3	60	45
4	70	1000

Progesterone penicillinase conjugate

Penicillinase was covalently coupled with 11-α-OH progesterone hemisuccinate using carbodiimide condensation reaction. For conjugation, enzyme and steroid were mixed a molar ratio of 1:30. After completion of condensation reaction, mixture was subjected to size exclusion chromatography using sephadex G-50 (15 x 300 mm) column. Data of results was shown in figure 3. Graph showed two main peaks between fraction number 9-14 and 26-29. Small peaks between fraction number 5-7 and 27- 29 were false peaks. From graph it was cleared that in first peak protein content was well in correlation progesterone concentration while in second peak it was not in correlation. Enzyme activity in first peak was found higher than second peak. According to gel exclusion chromatography principal, first peak might contain conjugate which was further confirmed by analyzing binding of conjugate with progesterone antibody on strip.

Optimization of progesterone penicillinase conjugates

Every enzyme shows its highest activity at an optimum pH. For obtaining an optimum pH, 10 µl of conjugate was used for iodometric titration at pH 4, 5.8, 7.0 and 8.0. Results are shown in figure 4 A and it confirmed that penicillinase enzyme activity of conjugate was highest at ph 4.0. Enzyme activities are also influenced by temperature. Each enzyme has its own optimum temperature. Data of results was shown in figure 4 B. Graph showed that optimum temperature for conjugate activity was 35 ºC.

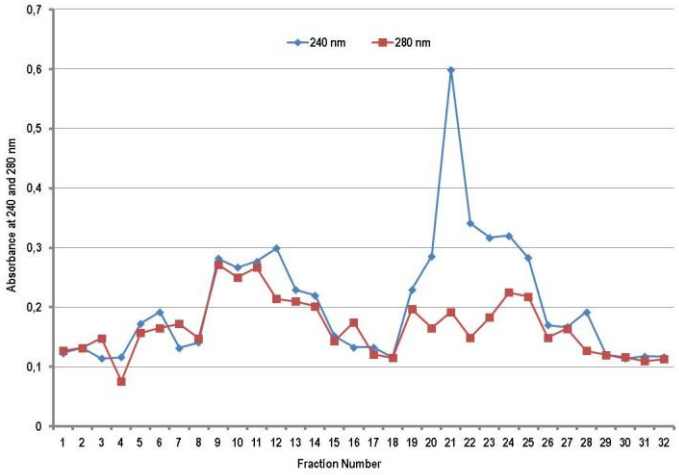


Figure 3 Elution profile of progesterone penicillinase conjugate in comparision with progesterone and protein content of eluted fractions.

Checkerboard analysis for standardization of antiserum and conjugate dilution for EIA

After characterization of antibody and enzyme conjugate, their optimum concentrations for use in dipsticks EIA was standardized using a checkerboard analysis. Nitrocellulose strips coated with different dilutions of progesterone antibody were dipped for 30 minutes in different dilutions of conjugate solutions. Standardization involved selection of a continuation of dilutions that would yield a visible score (decolourization of starch iodine solution) in about 10 minutes. Results, as shown in table 4, indicate that optimum progesterone antibody and conjugate dilution were 1:20 and 1:40 respectively, as at these dilution a visible score was perceived in about 10 minutes.

According to reports, pregnant animals maintain nearly 4-5 fold higher progesterone levels than non-pregnant animals around 18-24 days post service. During this period (18-24 days) a pregnant animal would have > 5 ng/ml progesterone while a non-pregnant animal < 1 ng/ml. Therefore, standardized progesterone antibody and conjugate dilution were titrated with standard progesterone concentration of 1 ng/ml and 5 ng/ml. Results were shown in table 5.

Optimum milk dilution

For optimizing milk dilution for pregnancy diagnosis, milk samples were collected from confirmed pregnant and non pregnant animals. They were

tested by already standardized antibody and conjugate dilution with their different dilutions of 1:1 to 1:5 in dilution buffer (0.1 M PBS buffer pH 7.2). Results from checker board analysis, as shown in table 6, showed that milk dilutions of 1:3 and 1:4 were optimum for EIA.

Validation of pregnancy diagnosis test

Twenty cow milk samples were collected from Livestock Research Center of University. In present study all animals were provided with natural service on their date of heat. Milk samples were suitably diluted with 0.1 M PBS dilution buffer pH 7.2. Nitrocellulose dipsticks coated with optimized dilution of antibody were then dipped in these diluted milk samples for one hour at room temperature. After washing and incubation with conjugate solution, dipsticks were placed in substrate solution for iodometric analysis. Out of 20 milk samples assayed by dipsticks EIA method, 7 cows were diagnosed as pregnant and 12 as non pregnant and one remained doubtful due to partial decolourization.

Table 4 Checker board analysis for standardization of antiserum and conjugate dilution for EIA. Dipsticks were coated with different antibody dilutions and exposed to different dilutions of conjugate for 30 minutes. Penicillinase activity was assayed by iodometric analysis. ++, -- and +- represent total decolourization, no decolourization and partial decolourization in 10 minutes respectively.

Antibody dilution	Conjugate dilution					
	1:20	1:40	1:80	1:160	1:320	1:640
1:20	++	++	+-	--	--	--
1:40	++	--	--	--	--	--
1:80	++	--	--	--	--	--
1:160	--	--	--	--	--	--
1:320	--	--	--	--	--	--
1:640	--	--	--	--	--	--

Data of results exhibited an excellent agreement between dipstick diagnosis and rectal palpation diagnosing 100 percent non pregnant animal while 87 percent for pregnant animal. One case of doubtful results was obtained by dipstick method.

Discussion

Many progesterone based immunodiagnostic procedures have been developed for pregnancy diagnosis in bovines. For a reliable immuno-pregnancy-test, a major constraint is unavailability of specific antibodies. Affinity purified progesterone antibodies were prepared for this purpose. Present investigation has demonstrated production of progesterone antibodies using 11-α-OH progesterone hemisuccinate conjugated BSA as an antigen and ability of affinity purification as a reproducible and reliable method for producing highly specific antibodies against progesterone. In this study progesterone antiserum, suitably diluted, was passed through an affinity column containing 6-amino hexanoic acid-sepharose 4B beads conjugated with progesterone carboxymethyl oxime. Antibodies bound to progesterone attached to beads while other proteins passed through column unabsorbed. It was essential to wash column till no protein was observed in eluent. It has been previously demonstrated that antibodies could be eluted by using a 5 to 2.5 pH gradient. A major peak had emerged at pH 4.0 showing low affinity antibodies. A most desirable faction was, however, fraction eluted with pH 2.7 buffer. It had high affinity for antigen. As low pH can damage native structure of antibody, the fractions were collected in 0.5 M Tris-HCL buffer pH 8.6. This treatment had neutralized excess acid and antibodies retained their native structures. Used affinity column contained a carboxymethyloxime bridge instead of a hemisuccinate bridge. Thus if antisera contained any antibodies which were specific to bridge (hemisuccinate), they would not bind onto beads. Thus, antibodies obtained from affinity column could be considered as specific for progesterone.

Another key reagent of kit was a solution of enzyme penicillinase and progesterone. For this purpose, progesterone hemisuccinate was conjugated with enzyme penicillinase using carboiimide condensation method. This mixture when passed through Sephadex G-50 size exclusion column, it could be resolved in peaks for enzyme- progesterone conjugate, free enzyme and free steroid. Elution profile was monitored at 280 nm for protein and 240 nm for steroid. Different fractions were collected and two major peaks were observed. First peak had emerged between fractions 9-11. This peak had absorption at 240 nm as well as 280 nm suggesting that it contained both steroid and protein. This was followed by a small minor peak at fraction 16 which had absorption at 280 nm but no at 240 nm. This could be suggested that it was due to free penicillinase which did not bind to progesterone. A major peak was also observed at fraction 19-25. In this region absorption at 240 nm was highest, indicating that this could be due to free progesterone. These possibilities were confirmed by enzyme analysis of fractions which showed that fractions 9-11 and 16 only had high enzyme activity and 19-25 had no enzyme activity at all. Additional confirmation was made by RIA (data not shown) where it was demonstrated that first peak (9-11) effectively bound to progesterone antibody.

Appropriate dilution of conjugate and antibody were standardized by checker board analysis for kit development. While making various standardizations it was observed that conjugate had exhibited maximum affinity at pH 7.0 and temperature of 35 ºC. Beside this it was found that penicilline solution was useable for nearly three weeks if it was stored in a well sealed bottle at 4 ºC. For effective decolourization of starch iodide solution, enzyme and iodine concentration optimization was a crucial parameter. This study demonstrated that concentration in range of 40-50 µM of iodine was decolorized in a reasonable time of 7-15 minutes with 0.04 units of enzyme.

Using these optimum conditions, 20 randomly selected milk samples were analyzed and excellent correlation was observed between pregnancy diagnosed by developed dipstick EIA method and conventional rectal palpation method. Remarkable, use of starch iodine and peniciline reduces sufficiently cost of kit.

In conclusion, this study presents a dipstick EIA kit containing low cost reagents and can be used successfully for pregnancy diagnosis.

Table 5 Titration of antibody dilutions i.e. 1:20, 1:40 and conjugate dilution 1:40.

Antibody dilution	Standard progesterone concentration	
	1 ng/ml	5 ng/ml
1:20	++	--
1:40	--	--

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